

REPORT ON CAPSTONE PROJECT

Module-End -1

Module End Assignment:

“Data Cleaning, Analysis & Visualization with Excel”

Dataset Link: E-commerce store dataset

Objective of the data set:

The dataset represents product sales based on categories from different stores in an e-commerce environment. It captures comprehensive information related to customers, products, stores, and sales transactions, enabling detailed analysis of purchasing behaviour, sales performance. 4 datasets were found in the e-commerce dataset.

Customer_Dim dataset describes Customer_Id, Customer_Name, age, Gender, City, Country, State and Loyalty level

Product_Dim dataset describes Product_Id, Name, Category, Sub_category, brand, cost, and stock

Store_Dim dataset describes the details of store_Id, Name, Region, City and flagship

Sales_Fact describes the details of Customers sales based on order date, quantity purchased, amount and payment_type

Data entry, import, and basic organization using tables:

After importing the data into excel, dataset should be converted into table for efficient data cleaning.

Before converting the range to table:

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	2024-07-07	CUST1409	PROD97	STORE3	1	218.04	0.09	Credit Card	198.4164
SALE1	2023-12-25	CUST1467	PROD74	STORE17	3	104.47	0.27	PayPal	228.7893
SALE2	2024-06-03	CUST1056	PROD49	STORE8	4	914.05	0.25	Credit Card	2742.15
SALE3	2023-11-29	CUST1346	PROD76	STORE8	1	921.51	0.16	COD	774.0684
SALE4	2024-01-06	CUST1493	PROD59	STORE5	3	745.3	0.18	Credit Card	1833.438
SALE5	2024-07-26	CUST1116	PROD15	STORE7	3	147.01	0.19	COD	357.2343
SALE6	2025-07-12	CUST1016	PROD56	STORE4	2	785.31	0.15	PayPal	1335.027
SALE7	2023-11-18	CUST1152	PROD70	STORE8	4	342.39	0.23	PayPal	1054.5612
SALE8	2025-05-04	CUST1187	PROD99	STORE5		448.25	0.07	PayPal	1250.6175
SALE9	2025-07-31	CUST1464	PROD33	STORE0	2	873.17	0.23	PayPal	1344.6818
SALE10	2024-02-13	CUST1065	PROD24	STORE14	1	278.63	0.07	Credit Card	259.1259
SALE11	2025-05-24	CUST1320	PROD39	STORE1	4	286.92	0.17	Credit Card	952.5744
SALE12	2024-08-24	CUST1237	PROD88	STORE4	4	644.58	0.01	Credit Card	2552.5368
SALE13	2025-08-06	CUST1443	PROD34	STORE16	2	316.9	0.3	PayPal	443.66
SALE14	2025-06-21	CUST1348	PROD24	STORE15	4	403.47	0.29	PayPal	1145.8548

Select the entire data set and click ctrl +T → entire Range will be converted to table

Sales_ID	Order_Dat	Customer_ID	Product_ID	Store_ID	Quantit	Unit_Pric	Discoun	Payment_Typ	Total_Amount
SALE0	2024-07-07	CUST1409	PROD97	STORE3	1	218.04	0.09	Credit Card	198.4164
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SALE14	2025-06-21	CUST1348	PROD24	STORE15	4	403.47	0.29	PayPal	1145.8548
SALE15	2024-12-10	CUST1435	PROD36	STORE9		871.34	0.04	COD	2509.4592
SALE16	2025-09-13	CUST1276	PROD65	STORE19	2	451.98	0.25	Credit Card	677.97
SALE17	2024-12-11	CUST1079	PROD27	STORE4	2	502.69	0.23	Credit Card	774.1426
SALE18	2024-08-13	CUST1094	PROD39	STORE4	2	559.23	0.25	COD	838.845
SALE19	2024-03-18	CUST1239	PROD60	STORE13	4		0.04	COD	1782.4512

Once dataset has been changed to the table, basic organization should be done.

Basic Organization includes changing the data font, size, alignment and table design



Data cleaning: find/remove duplicates, handle missing/inconsistent data, correct formatting errors, and validate data:

Once data organization has been done, data cleaning should be performed.

Steps followed in cleaning the data:

- 1) All columns were trimmed to eliminate both preceding (leading) and trailing spaces, ensuring data consistency

Formula: =TRIM(C2)

(Or)

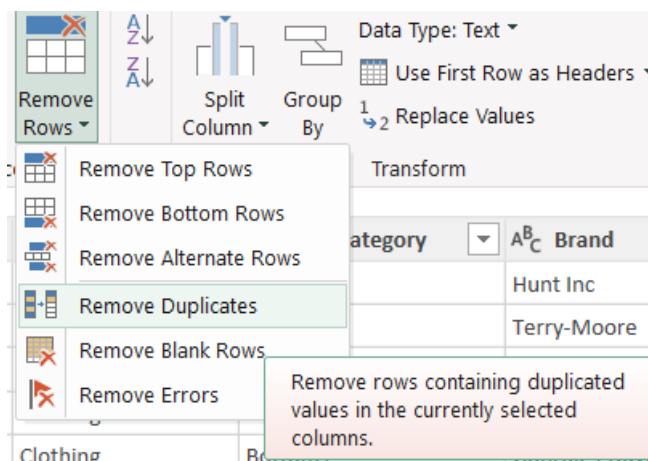
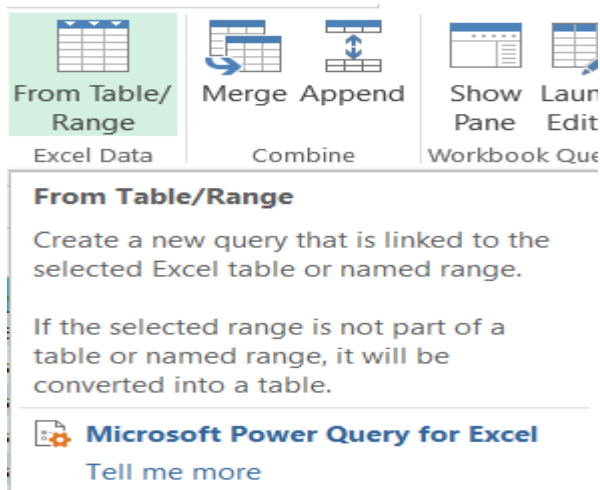
- 2) Transform the Data to power query → Transform → format → select the entire data → Trim

Removing duplicate data:

Removed duplicate data from the dataset to maintain data accuracy.

Imported the data into power query to eliminate the duplicate records.

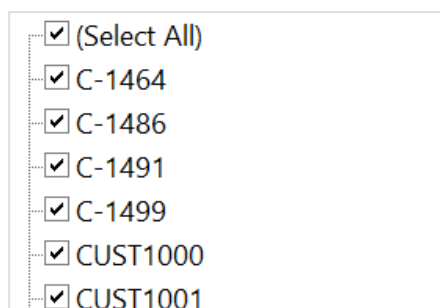
Goto PowerQuery Ribbon → select from table/Range option → click ok



Once the data was loaded into Power Query, duplicates were eliminated using a primary key consisting of Customer_ID, Product_ID, Store_ID, and Sales_ID.

Handle missing/inconsistent data:-

In Customer_Dim dataset, inconsistent data were found in the column Customer_Id.



C-1464 should be CUST1464. To replace this data, applied find and replace command has been used.

Goto → Home → Editing Ribbon → select Find & Select → select Replace option

Data Inconsistent found in the Country Column. To make it as consistent, all data should be changed to same value

Text Filters

Search

☒ (Select All)
☒ United States of America
☒ USA

Search

☒ (Select All)
☒ USA

Customer Name contains the title for some values. It should be removed

Amber Jackson	Amber Jackson
Daniel Boyd	Daniel Boyd
Melissa Howard	Melissa Howard
Ms. Maria Leach	Maria Leach
Dominic Page	Dominic Page
Brian Lopez	Brian Lopez

Null values were found in the column Loyalty_level.

To handle the null values, **mode** has been calculated based on the **Loyalty Level** and applied

Load the data into power query → Goto transform ribbon → group by has to be selected → and count aggregate function performed based on the Loyalty Level column

Group By

☒ Basic
 ☐ Advanced

Specify the column to group by and the desired output.

Group by

Loyalty_Level

New column name

Count

Operation

Count Rows

Column

	AB C Loyalty_Level	1.2 Count
1		4
2	Platinum	145
3	Bronze	125
4	Silver	104
5	Gold	122

So, after taking group by, platinum has the highest frequency. Platinum is replaced in the place of null values.

Before

Loyalty_Level
Platinum
Platinum
Bronze
Silver
Bronze
Platinum
Bronze
Gold

After

Loyalty_Level
Platinum
Platinum
Platinum
Bronze
Silver
Bronze
Platinum
Bronze
Gold
Platinum

Product dim Dataset Cleaning:

Inconsistent data was found in the product id column. So, find and Replace has been applied to make it as consistent.

Product_ID
P-3
P-16

Product_ID
PROD3
PROD16

Null values are found in the cost column. So, to fill the values, mean has been taken based on Sub_Category and applied.

Formula : =IF([@Cost] =
"",AVERAGEIF([Sub_Category],[@[Sub_Category]],[Cost]),[@Cost])

Cost
112.42
342.56
364.67
306.46
261
312.8

Cost
112.42
342.56
296.83
364.67
306.46
261.00
312.80
218.86

Null values are found in the Stock column. So, to fill the values, mean has been taken based on Sub_Category and applied.

Formula:

=IF([@Stock]="",AVERAGEIF([Sub_Category],[@[Sub_Category]],[Stock]),[@Stock])

Stock ▾
416
181
352
452
487
59
350
170
142
323
308

Stock ▾
416
181
352
452
487
59
350
170
142
323
308
270

Null values are found in the columns quantity, Total_Amount, Unit_Price and discount, since these values are numerical data, average can be taken based on quantity, Total_Amount, Unit_Price and discount column and replaced in the null values

Formula: =IF([@Quantity] = "", AVERAGE ([@Quantity]),[@Quantity])

1
3
3
2
4
2
1

1
3
3
2
4
2
2
1

Formula: =@IF([@[Unit_Price]] = "", AVERAGEIF([Store_ID],[@[Store_ID]],[Unit_Price]),[Unit_Price])

Unit_Price ▾

Unit_Price ▾
569.50
521.79
505.34

Formula:

=@IF([@Discount]="",AVERAGEIF([Store_ID],[@[Store_ID]],[Discount]),[Discount])

Discount	Discount
	0.140079365
	0.140285714
	0.153551402
	0.142456609

Formula:

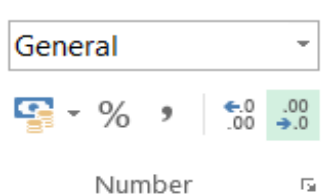
=@IF([@[Total_Amount]]="",AVERAGEIF([Store_ID],[@[Store_ID]],[Total_Amount]),[Total_Amount])

Total_Amount	Total_Amount
	1128.811844
	989.56525
	1111.27738
	1001.057962

Correct formatting errors:

Applied formatting to reduce the decimal in the Amount column.

Click Home → Number Ribbon → decrease decimal



Total_Amount ▾
198.4164
228.7893
2742.15
774.0684
1833.438
357.2343
1335.027
1054.5612
1250.6175

Total_Amount ▾
198.42
228.79
2742.15
774.07
1833.44
357.23
1335.03
1054.56
1250.62

Remove the column that contains irrelevant data if it exists

Based on the analysis, all columns contain relevant and meaningful data, so no columns were removed.

Application of Excel formulas for calculations, summaries, and logical/text operations

SUM, IF, COUNTIF, TEXT, Average, xlookup/vlookup functions.

Sum and Average can be applied only on numerical data. So calculated sum and average for the numerical columns such as Cost and Stock

Formula: =SUM(F2:F101)

Formula: =SUM(G2:G101)

Average Formula: =AVERAGE(F2:F101)

Countif Formula= =COUNTIF(D2:D101,"Appliances")

SUM =	25529.36	27400
Average =	255.29	274
Count If sub category based on appliances=	3	

Text functions such as TRIM and PROPER has performed to remove leading and trailing spaces and format the text to proper case

Formula: =TRIM(C2)

PROPER(B2:B501)

Count of Customer Id = =CountA(A2:A501)

Count of Customer Id =	500
------------------------	-----

SUM	1027475.74			2173476.01
AVERAGE	513.74			1086.74

In Order_date, timestamp is there in the data which is not required. So, it can be removed.

 Accounting
Order_Date

 Short Date
Order_Date

Order_Date ▾
07-07-2024 00:00
25-12-2023 00:00
03-06-2024 00:00
29-11-2023 00:00
06-01-2024 00:00
26-07-2024 00:00
12-07-2025 00:00
18-11-2023 00:00

Order_Date ▾
07-07-2024
25-12-2023
03-06-2024
29-11-2023
06-01-2024
26-07-2024
12-07-2025
18-11-2023

Flash Fill:

In the Sales_Fact table, year column has been created for year-based analysis. So, I have used flash fill option to extract the year from the date.

Order_Year ▾
2024
2023
2024
2023
2024
2024
2025
2023
2025
2025

VLOOKUP:

By using vlookup, searched the city of the particular customer based on the cust_name.

=VLOOKUP(B19,Table1__3[[Name]:[Loyalty_Level]],4,0)

Combining tables using VLOOKUP:

=VLOOKUP(\$A2,Table1__3,2,0)

=IFERROR(VLOOKUP([@[Customer_ID]],Table4_1__24[[Customer_ID]:[Order_Y ear]],4,FALSE),0)

=SUMIFS(Table4_1__24[Quantity],Table4_1__24[Customer_ID],[@[Customer_ID]])

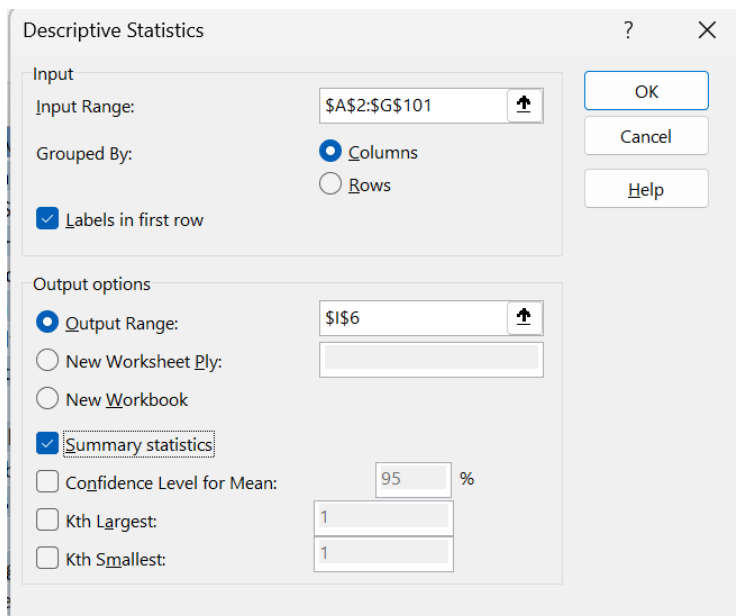
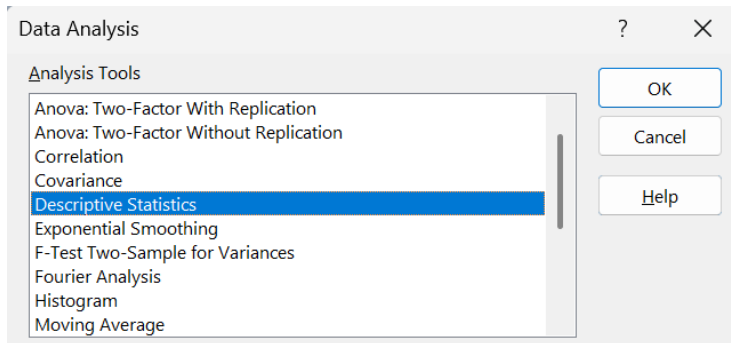
=SUMIFS(Table4_1__24[Total_Amount],Table4_1__24[Customer_ID],[@[Customer_ID]])

Customer_ID ▾	Cust_Name ▾	No.of Products Purchased ▾	Total Amount ▾
CUST1000	Sheena Bonilla	6	2437.67
CUST1001	Eugene Nelson	5	1941.19
CUST1002	Benjamin Price	7	2078.42
CUST1003	Jeremy Harris	9	2330.87
CUST1004	Caroline Collins	13	5830.65
CUST1005	Christine Vaughn	13	4795.10
CUST1006	Amber Jackson	12	6348.02
CUST1007	Daniel Boyd	25	10211.02
CUST1008	Melissa Howard	6	4029.71
CUST1009	Maria Leach	7	3903.61
CUST1010	Dominic Page	5	2235.55
CUST1011	Brian Lopez	10	3868.39
CUST1012	Kenneth Lambert	17	7752.78
CUST1013	Nathan Hernandez	24	13005.77
CUST1014	Ashley Cohen	4	2365.20
CUST1015	Laura Alvarez	14	5610.02
CUST1016	John Scott	15	5070.84
CUST1017	Adam Cruz	14	3968.96
CUST1018	Rhonda Moss	5	1963.97
CUST1019	Michael Torres	10	3751.71

XLOOKUP:

=XLOOKUP("James Berger",Table1__3[Name],Table1__3[Loyalty_Level],"No Records")

Descriptive statistics (mean, median, mode, standard deviation, etc.) using formulas or Analysis ToolPak



Analysis Toolpak	
Mean	1086.74
Standard Error	18.30
Median	873.17
Mode	989.57
Standard Deviation	818.51
Sample Variance	669957.47
Range	3889.78
Minimum	18.46
Maximum	3908.24
Sum	2173476.01
Count	2000

Use of Pivot Tables for advanced summarization:

Pivot table is used to summarize, analyze, and present data from a table in a structured way.

Row Labels	Count of Customer_ID
Female	262
Male	238
Grand Total	500

Sum of Total_Amount	Payment Type			
Year	COD	Credit Card	PayPal	Grand Total
2023	67866.54	82595.15	72065.26	222526.95
2024	358305.66	319429.08	416050.78	1093785.52
2025	277453.93	310987.63	268721.98	857163.54
Grand Total	703626.13	713011.86	756838.02	2173476.01

Store_Id	Average of Total_Amount	Sum of Qua
STORE0	915.47	204
STORE1	1075.11	231
STORE10	1128.81	255
STORE11	1057.37	206
STORE12	1168.29	259
STORE13	1204.23	243
STORE14	1118.40	224
STORE15	1049.25	243

Order_Year

2023

2024

2025

Payment_Type

COD

Credit Card

PayPal

Sub_Category	Sum of Stock
Accessories	4585
Appliances	596
Audio	965
Bottoms	1179
Computer	1795
Decor	2319
Equipment	1320
Fitness	2410
Footwear	792
Fragrance	711
Furniture	338
Grooming	256
Haircare	1419
Kitchen	651
Lighting	1188
Makeup	1006
Mobile	301
Outdoor	734
Outerwear	1243
Skincare	1424
Storage	165

The dashboard summarizes sales and operational data using multiple pivot tables. It shows customer distribution by gender, Stocks based on sub_category

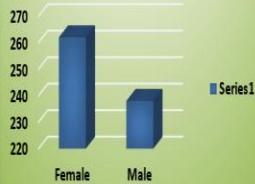
Year-wise sales analysis indicates that 2024 recorded the highest total sales, followed by 2025 and 2023. Among payment methods, PayPal contributes the highest transaction value, followed by credit cards and cash on delivery.

Slicers for year, and payment type enable interactive filtering and focused analysis of the data.

Charts and Dashboard:

E-COMMERCE SALES & INVENTORY PERFORMANCE DASHBOARD

GENDER-WISE CUSTOMER DISTRIBUTION



Order...

2023

2024

2025

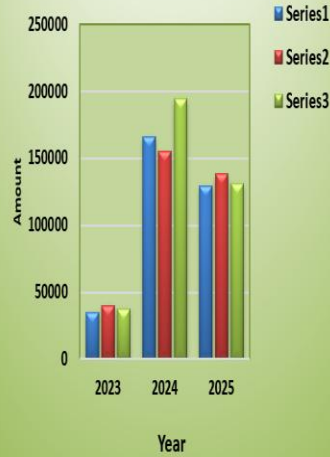
Payme...

COD

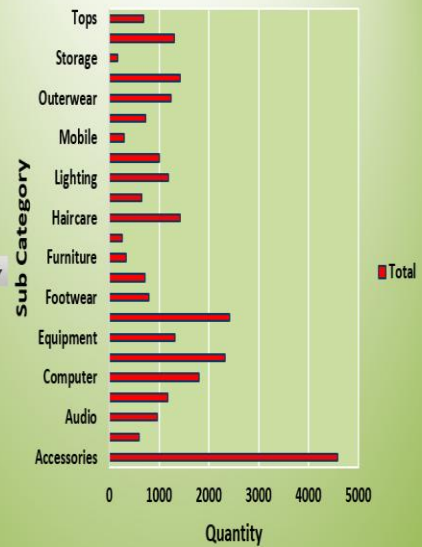
Credit Card

PayPal

Transactions by Payment Type (Year-wise)

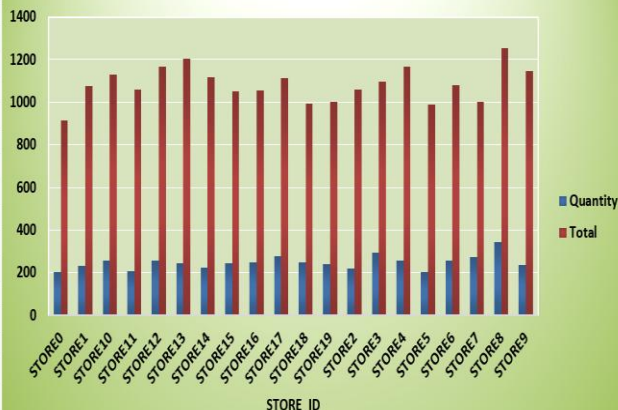


Stock by Sub-Category



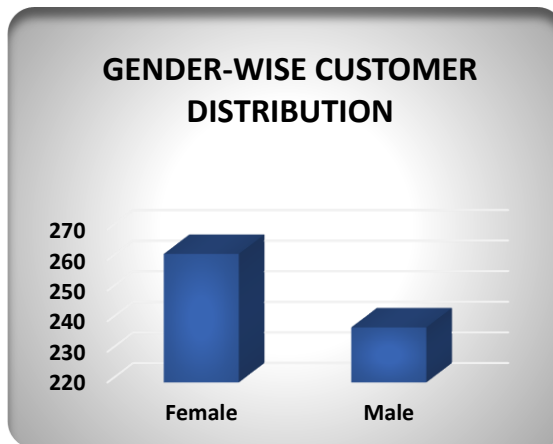
E-COMMERCE SALES & INVENTORY PERFORMANCE DASHBOARD

Sales by Store



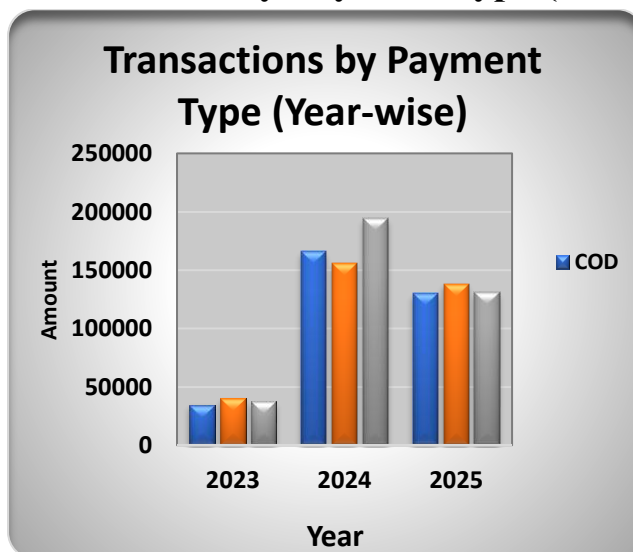
Dashboard Summary

1. Gender-wise Customer Distribution



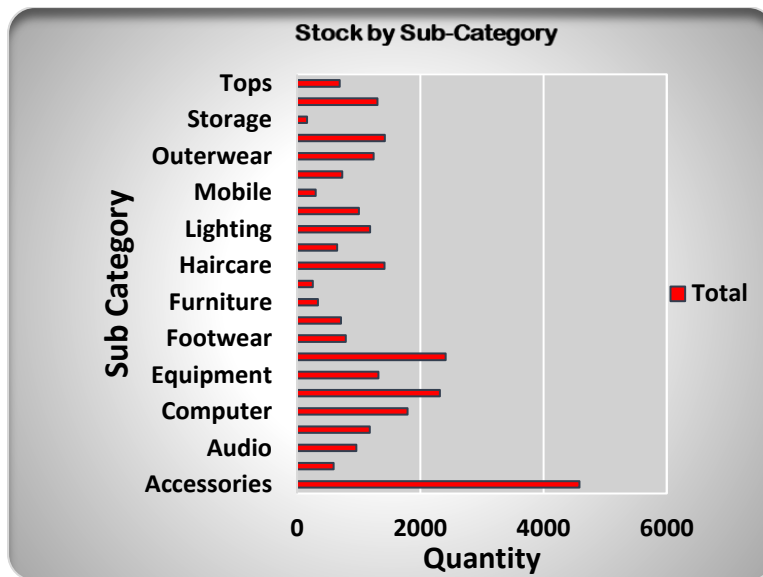
- Female customers are higher than male customers.
- Female count is approximately 260+, while male count is around 240+.
- This indicates stronger engagement or purchasing activity from female customers.

2. Transactions by Payment Type (Year-wise)



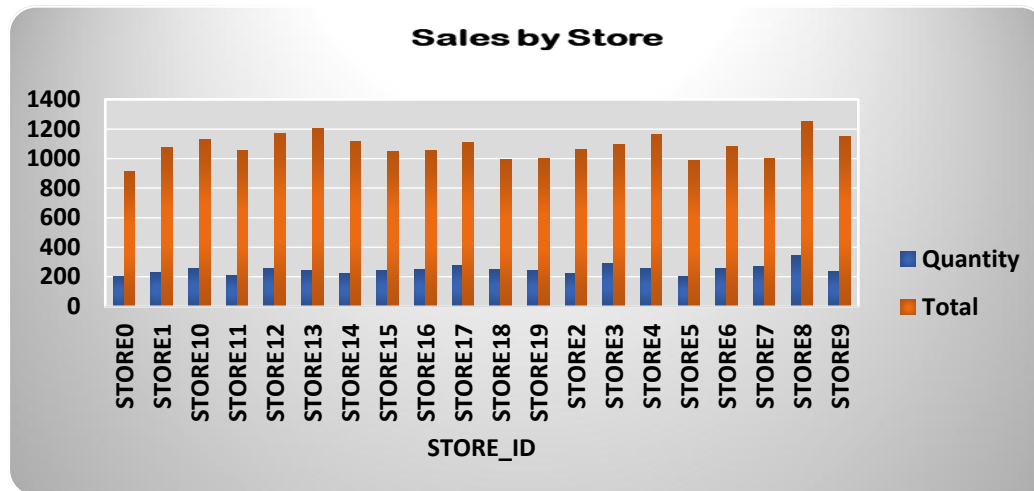
- Overall transactions increased significantly in 2024 compared to 2023.
- Series3 shows the highest transaction amount in 2024, reaching close to 200K.
- Transaction values slightly declined/stabilized in 2025 compared to 2024.
- Payment methods are fairly balanced, but one payment type dominates in each year.

3. Stock by Sub-Category



- Accessories has the highest stock quantity, exceeding 4500 units.
- Other highly stocked categories:
 - Equipment
 - Computer
 - Haircare
- Categories with relatively low stock:
 - Furniture
 - Mobile
 - Storage
- Inventory distribution suggests higher demand or priority for electronics and accessories.

Sales by Store – Summary:



- The dashboard presents store-wise comparison of **Quantity Sold** and **Total Sales**.
- **STORE8** recorded the highest sales performance with total sales exceeding **1200**.
- **STORE0** showed the lowest sales among all stores.
- Quantity sold across stores remains fairly consistent, while total sales vary significantly.
- Stores such as **STORE3**, **STORE12**, **STORE13**, and **STORE8** performed strongly in overall sales.
- The variation between quantity and total sales indicates differences in product pricing or transaction value across stores.

Insights:

- The business experienced strong growth in transactions during 2024.
 - Female customers form the larger customer base.
 - Inventory is heavily concentrated in accessories and electronic-related categories.
 - Some low-stock categories may require replenishment planning depending on demand trends.
 - Certain stores are generating higher revenue without a major increase in quantity sold.
 - High-performing stores can be analyzed further to identify successful sales strategies.
- Lower-performing stores may require:
- promotional campaigns,
 - inventory optimization,

- or pricing improvements.

*******End of the Document*******